BACULA BACKUP SERVER

Installation, Configuration, Backup Validation and Windows Recovery Lab

Technical implementation record | British English | 8 October 2026

VirtualBox · Debian 13 · Bacula 15.0.3 · PostgreSQL 17 · Synology NAS · Linux and Windows clients · Nagios integration

Document scope and evidence

This document records the lab configuration and outcomes discussed and verified during implementation. It is a reproducible technical guide, not an assertion that every installation command was captured verbatim. Commands and resource blocks below should be reviewed against the live systems before reuse. Credentials are deliberately replaced by placeholders.
This edition includes laboratory screenshots documenting installation, configuration, backup, recovery and monitoring. Screenshots are presented as evidence of the actual lab activities.

1. Lab architecture

System	IP address	Role
BACULA-SERVER (Debian VM)	X.X.X.118	Director, Storage Daemon, File Daemon, PostgreSQL catalogue; NFS mount /backup
HP Z400 (Linux physical host)	X.X.X.201	Bacula File Daemon; /XXX/XXXX backup source
Windows 10 physical host	X.X.X.184	DESKTOP-G07AFCU; Bacula File Daemon; user data
Synology NAS	X.X.X.145	NFS export /volume1/BaculaBackups
NAGIOS-SERVER (Debian VM)	X.X.X.83	NRPE checks for backup success/failure

Data flow: Director (BACULA-SERVER) schedules jobs → client File Daemons read source files → Storage Daemon writes Bacula volumes into /backup → NFS forwards writes to Synology NAS. PostgreSQL maintains job and file catalogue metadata. Nagios queries the catalogue through NRPE on BACULA-SERVER.

2. Baseline and services

Verified installed Bacula version: 15.0.3-3. The Director, Storage Daemon and File Daemon run on BACULA-SERVER. The catalogue uses PostgreSQL 17. The Director listens on 127.0.0.1:9101; this is valid because the Director initiates connections to the Windows File Daemon on TCP 9102. Windows does not need inbound access to Director port 9101.

2.1 Service and console checks

Run on BACULA-SERVER (X.X.X.118):

```
systemctl status bacula-director bacula-sd bacula-fd --no-pager
```

```
systemctl status postgresql --no-pager
/usr/sbin/bacula-dir -t -c /etc/bacula/bacula-dir.conf
bconsole
```

Troubleshooting note: on this Debian installation, bacula-dir was not in the interactive shell PATH; /usr/sbin/bacula-dir is the working executable path. Do not restart the Director unless the configuration test succeeds.

3. Synology NAS storage

The NAS exports X.X.X.145:/volume1/BaculaBackups via NFS; BACULA-SERVER mounts this export at /backup. Bacula File storage uses volumes on that mount. A verified check showed approximately 5.4 TB total, 4.2 TB used, 1.3 TB available (78% utilisation).

Run on BACULA-SERVER:

```
findmnt /backup
df -h /backup
mount | grep " /backup "
ls -lh /backup
```

At the time of the Windows Full backup, Vol-0008 was approximately 23 GB and its modification time matched the backup completion window. This provides storage-side corroboration; the authoritative backup outcome remains the Bacula job record. Before production use, check persistent /etc/fstab mounting, permissions, NAS availability at boot, and capacity alerting.

4. Linux client: HP Z400

The existing Bacula client is XXXX-HP-Z400-Workstation-fd. The FileSet Z400-Home includes /home/XXX while excluding VirtualBox VM directories, avoiding unnecessary virtual disk backups.

```
FileSet {
  Name = "Z400-Home"
  Include {
    Options { signature = MD5 }
    File = "/home/labadmin"
  }
  Exclude {
    File = "/home/labadmin/VirtualBox VMs"
    File = "/home/labadmin/VirtualBoxVMs"
  }
}
```

The Backup-Z400 job uses the existing DefaultJob definitions, File storage and File pool. WeeklyCycle schedule observed: Full on the first Sunday at 23:05, Differential on other Sundays at 23:05, and Incremental Monday–Saturday at 23:05. Validate the actual live schedule before changing it.

4.1 Linux backup and restore test

A test file was created, captured in an incremental backup, deleted, restored and checked successfully in the earlier lab. A later Full job (JobId 42) completed with 170,124 files and 181,500,372,532 bytes on 5 October 2026. Retain job logs and screenshots for the portfolio appendix.

5. Windows 10 client installation and connectivity

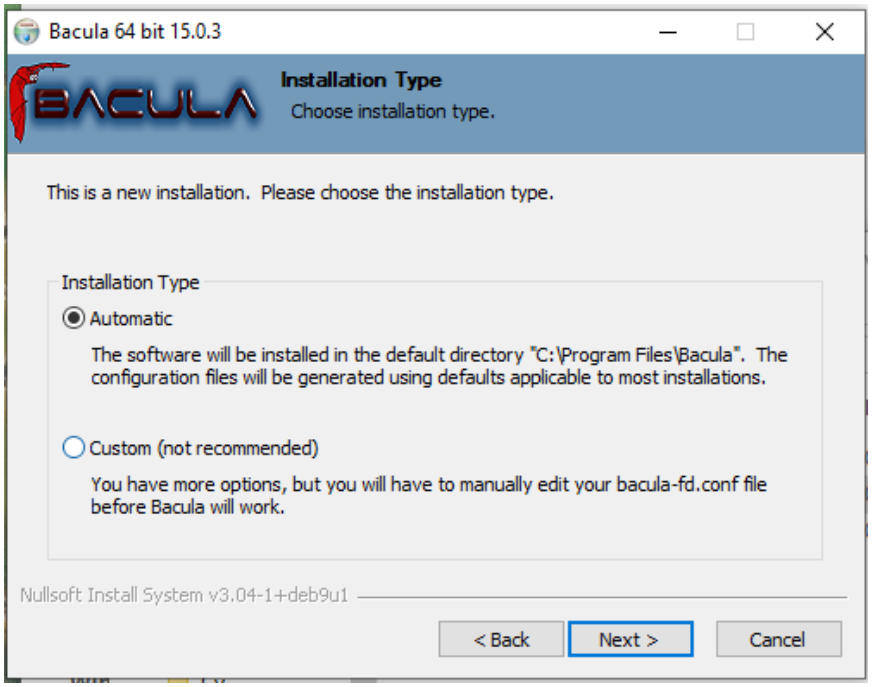


Figure 1. Bacula 15.0.3 Windows client installer: automatic installation selected.

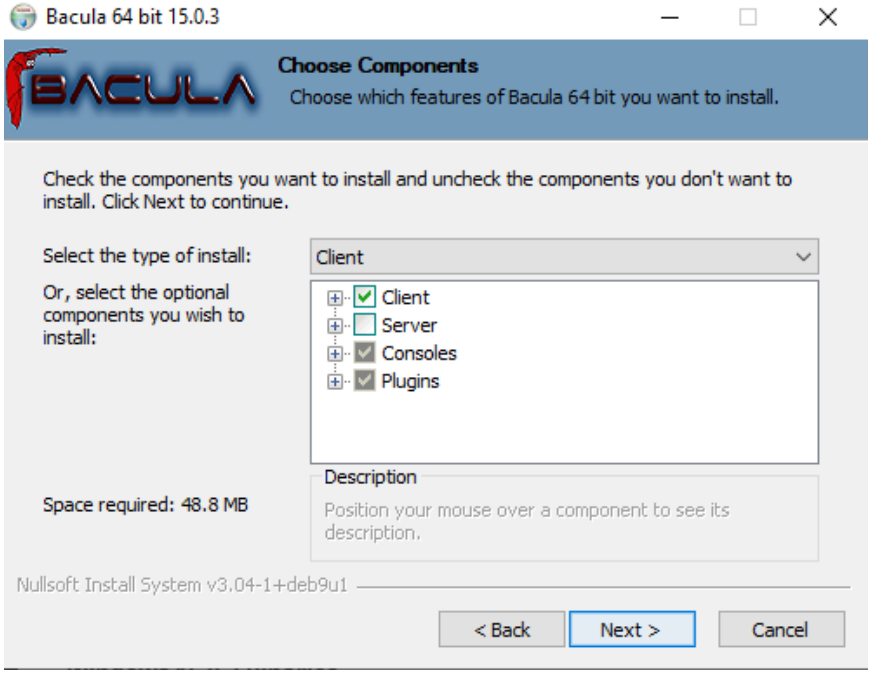


Figure 2. Bacula installer components: Client selected rather than the Server role.

Windows host: DESKTOP-G07AFCU (X.X.X.184). Bacula Windows File Daemon 15.0.3 64-bit installed. Configuration path: C:\Program Files\Bacula\bacula-fd.conf. Service: Bacula-fd (Automatic, Running), listening on TCP 9102.

5.1 Windows File Daemon identity

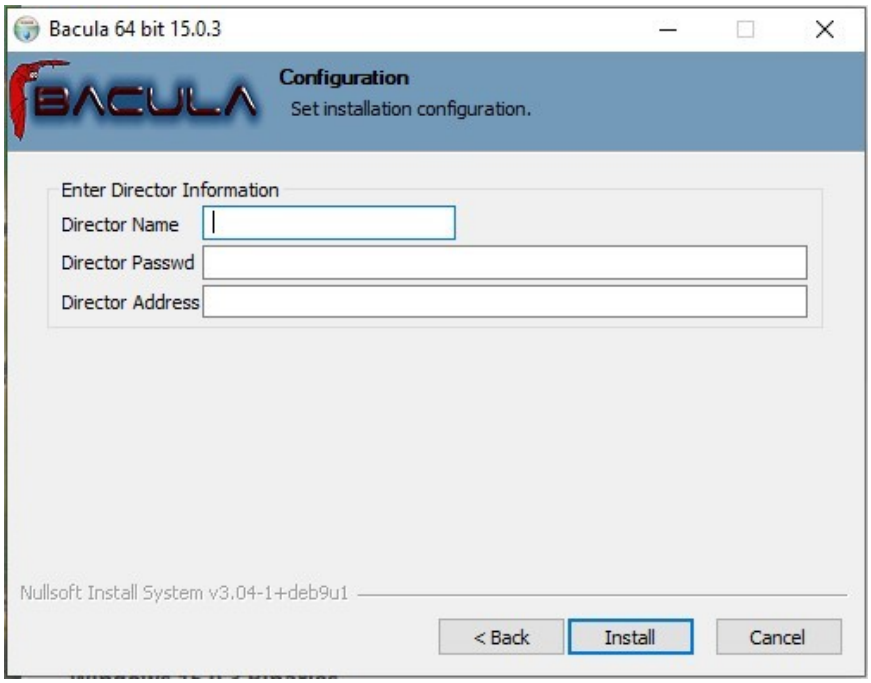


Figure 3. Director connection fields in the Windows installer. Use the authorised Director name and matching secret; never publish the password.

In bacula-fd.conf, the authorised Director block must use Name = bacula-server-dir and the password matching the Director Client resource. FileDaemon Name is desktop-fd. The Messages director reference also uses bacula-server-dir. An additional monitor Director block may remain present.

```
Director {
  Name = bacula-server-dir
  Password = "<WINDOWS_FD_PASSWORD>"
}
FileDaemon {
  Name = desktop-fd
  FDport = 9102
}
# Review all other existing bacula-fd.conf directives before replacing blocks.
```

Do not replace the entire Windows configuration with this excerpt. It illustrates only the key matching identities and redacted credential.

5.2 Restrict the Windows firewall

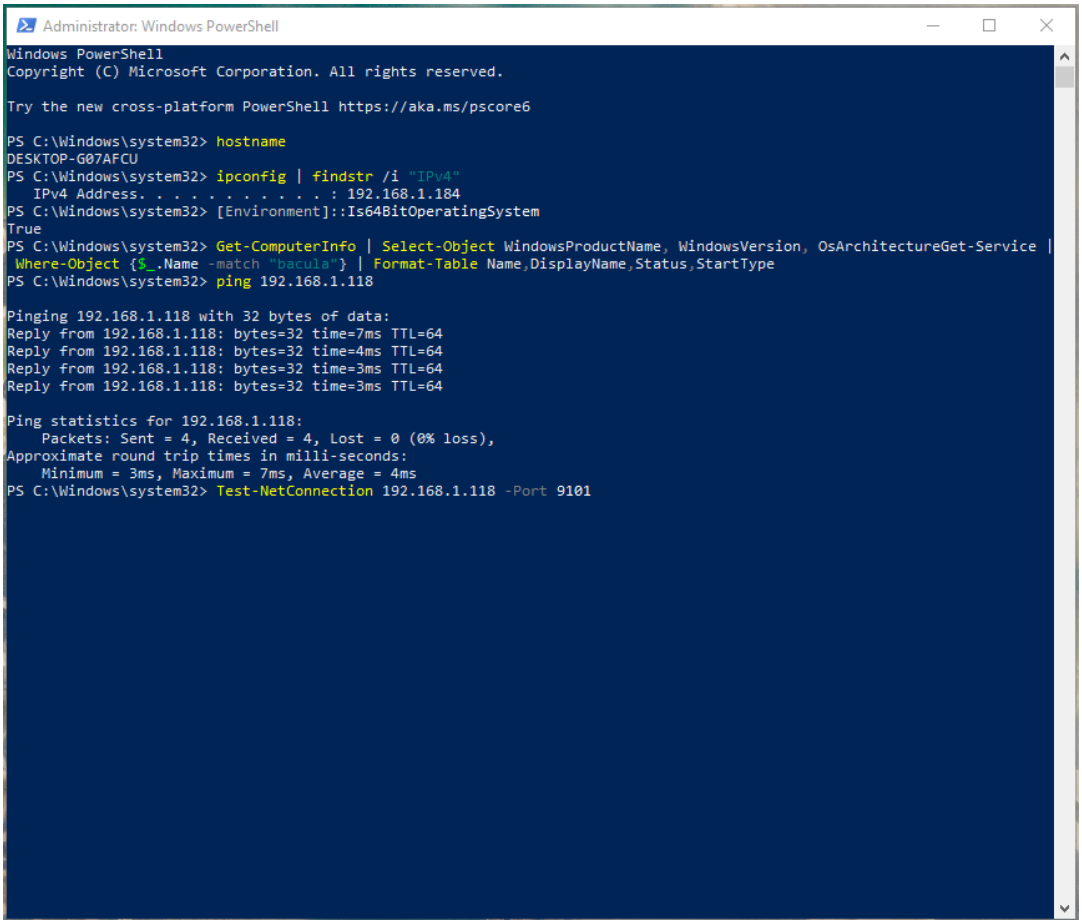


Figure 4. Windows PowerShell: host identity, IP address and connectivity diagnostics. Director port 9101 is not required inbound from Windows for scheduled backups.

Run in elevated PowerShell on WINDOWS 10 (X.X.X.184):

```
New-NetFirewallRule `
 -DisplayName "Bacula File Daemon - Bacula Server" `
 -Direction Inbound -Protocol TCP -LocalPort 9102 `
 -RemoteAddress X.X.X.118 -Action Allow
```

Run on BACULA-SERVER to validate TCP connectivity:

```
nc -zv -w 3 X.X.X.184 9102
```

Observed result: TCP 9102 open. Initial failure was caused by the Windows firewall. A later TLS handshake error was resolved by matching the File Daemon Director password and Director name; no TLS downgrade was needed.

6. Bacula Director: Windows client, FileSet and Job

Edit /etc/bacula/bacula-dir.conf on BACULA-SERVER, after making a dated backup. These are the working resource definitions used in the lab (password redacted).

```
cp -a /etc/bacula/bacula-dir.conf /etc/bacula/bacula-dir.conf.bak-before-windows-client
```

```
Client {
  Name = desktop-fd
  Address = X.X.X.184
  FDPort = 9102
  Catalog = MyCatalog
  Password = "<WINDOWS_FD_PASSWORD>"
  File Retention = 60 days
  Job Retention = 6 months
  AutoPrune = yes
}

FileSet {
  Name = "Windows-User-Data"
  Include {
    Options { signature = MD5 }
    File = "C:/Users/LabUser/Desktop"
    File = "C:/Users/LabUser/Documents"
    File = "C:/Users/LabUser/Downloads"
    File = "C:/Users/LabUser/Pictures"
  }
}

Job {
  Name = "Backup-Windows"
  JobDefs = "DefaultJob"
  Client = desktop-fd
  FileSet = "Windows-User-Data"
}
```

The inherited DefaultJob settings include the existing WeeklyCycle schedule, File1 storage and File pool. Inspect JobDefs before applying to a different installation.

Run on BACULA-SERVER:

```
/usr/sbin/bacula-dir -t -c /etc/bacula/bacula-dir.conf
systemctl restart bacula-director
bconsole
```

Inside bconsole on BACULA-SERVER:

```
status client=desktop-fd
list jobs
```

7. Windows backup execution and verified outcomes

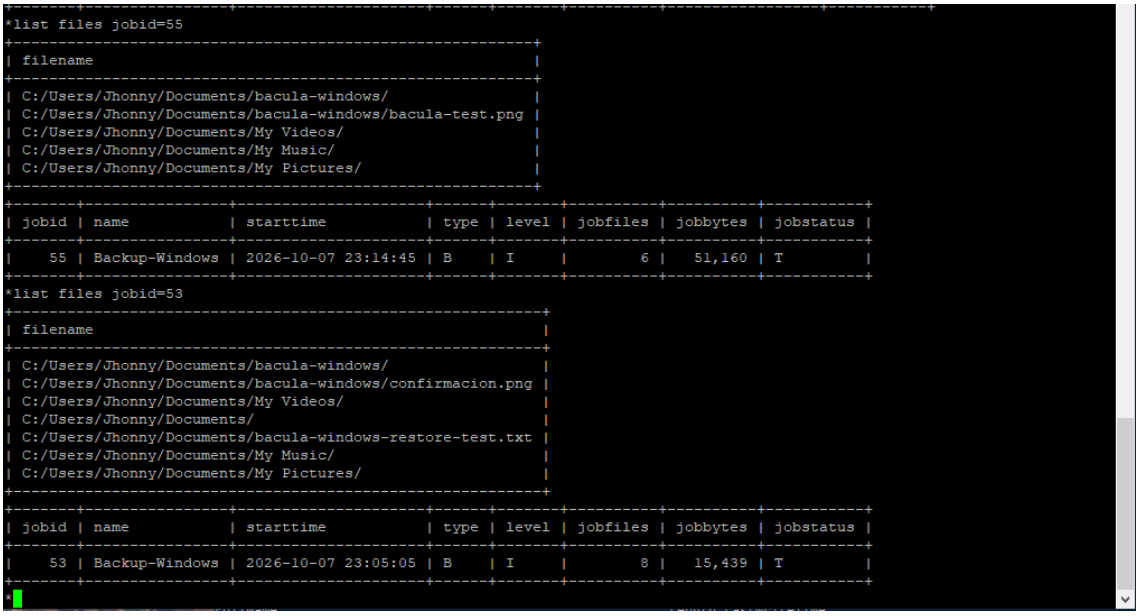


Figure 5. Bacula catalogue: successful incremental Windows backup jobs 53 and 55, including the restore test file.

JobId	Operation	Verified result
50	Windows Full backup	T (success); 550 files; 26,833,571,540 bytes; 7 Oct 2026, 14:52–15:15
53	Scheduled Windows Incremental	T; 8 files; 15,439 bytes; test file included
55	Manual Windows Incremental	T; 6 files; 51,160 bytes; 7 Oct 2026 at approximately 23:15
56	Windows restore	T; 1 file; 372 bytes

Bacula catalogue status T indicates normal job termination. For operational assurance, verify both job messages and storage availability, not merely the last successful timestamp.

8. End-to-end Windows recovery validation

8.1 Create the test file and record its checksum

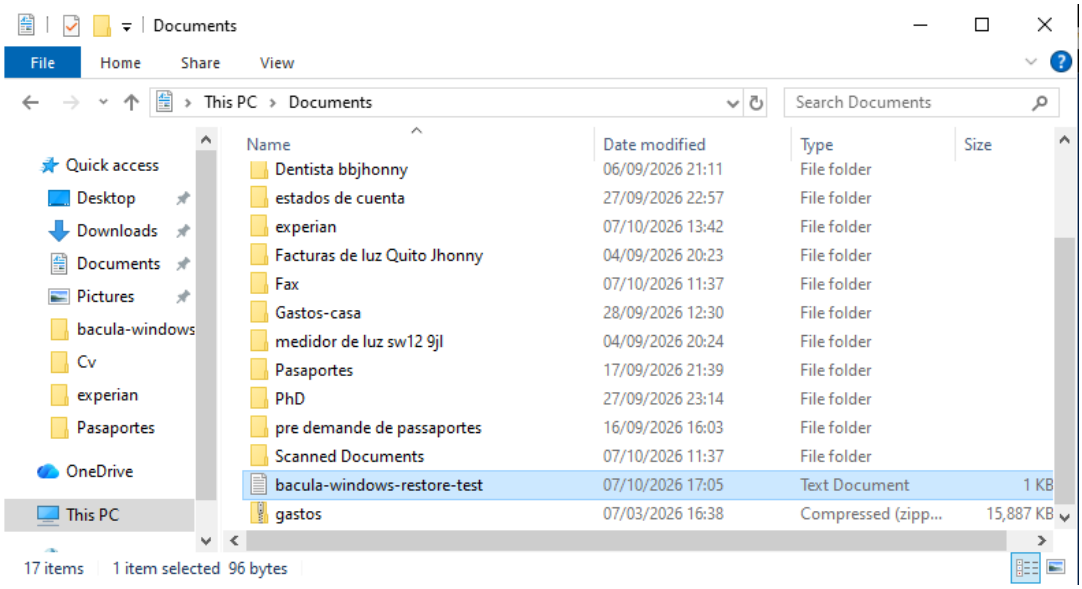


Figure 6. Windows Documents folder containing the test file before deletion.

Run in PowerShell on WINDOWS 10:

```
"BACULA WINDOWS RESTORE TEST - 07 OCTOBER 2026" |
  Out-File "$env:USERPROFILE\Documents\bacula-windows-restore-test.txt"
Get-Content "$env:USERPROFILE\Documents\bacula-windows-restore-test.txt"
Get-FileHash "$env:USERPROFILE\Documents\bacula-windows-restore-test.txt" -Algorithm SHA256
```

The test file was included in incremental JobId 53. The original pre-deletion SHA256 was not available for comparison in the final evidence; therefore cryptographic identity to the pre-deletion file remains unverified.

8.2 Delete the test file

Run on WINDOWS 10, only after confirming a completed backup:

```
Remove-Item "C:\Users\LabUser\Documents\bacula-windows-restore-test.txt"
Test-Path "C:\Users\LabUser\Documents\bacula-windows-restore-test.txt"
```

Observed result: False.

8.3 Restore through bconsole

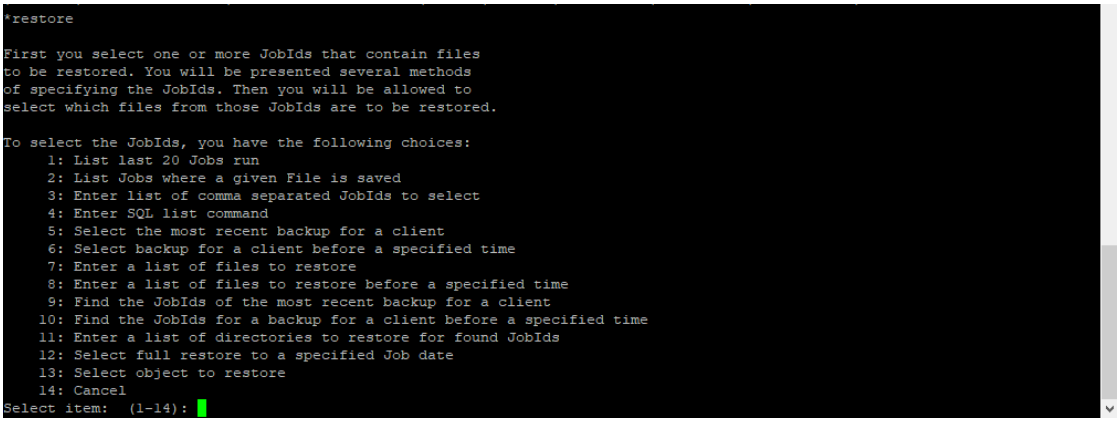


Figure 7. Bacula bconsole restore menu, selecting backup JobId 53.

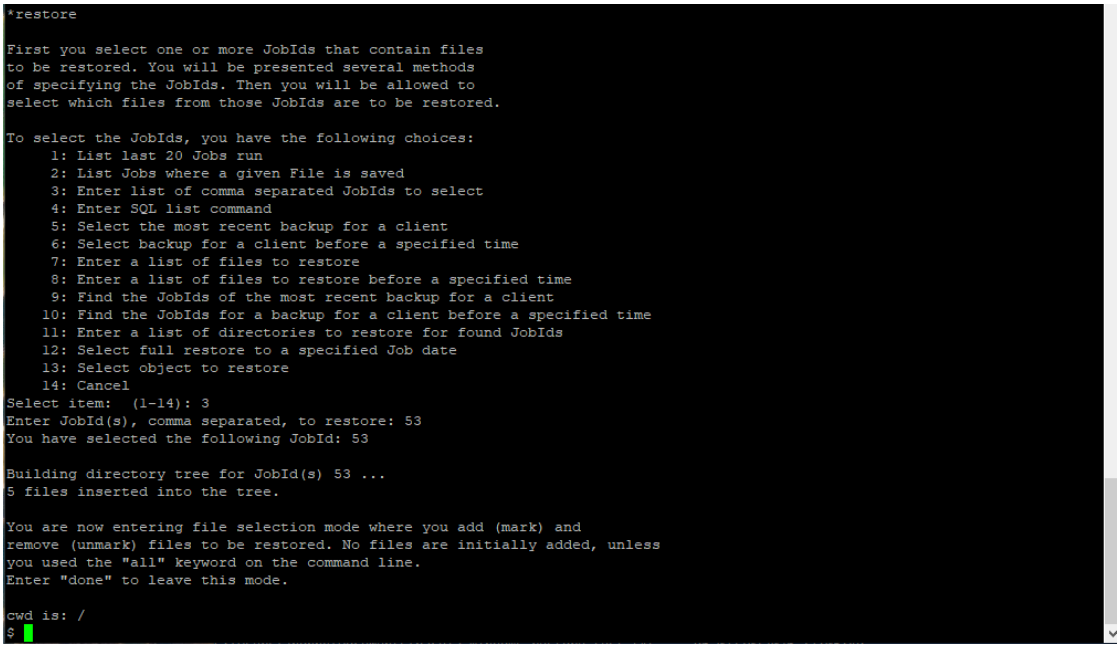


Figure 8. Restore file-selection tree created from the selected incremental job.

Run on BACULA-SERVER within bconsole:

```
restore
```

Select menu option 3 (JobIds) and enter 53. In the restore tree navigate to C:/Users/LabUser/Documents, mark bacula-windows-restore-test.txt, inspect lsmark and enter done. Select restore client desktop-fd (option 2 in the observed menu). At the confirmation screen choose mod → Where (option 9 in the observed menu) → / to remove the default invalid restore prefix. Confirm that Where is blank, only one file is selected, and the correct restore client is shown; then answer yes. Menu numbers may vary between Bacula versions.

The default Where value was /nonexistent/path/to/file/archive/dir/bacula-restores and was unsuitable. Restore used Vol-0008 and File1. Replace was Always, but only the deleted test file was selected.

8.4 Verify the restored file

Run on WINDOWS 10:

```
Test-Path "C:\Users\LabUser\Documents\bacula-windows-restore-test.txt"
Get-Content "C:\Users\LabUser\Documents\bacula-windows-restore-test.txt"
Get-FileHash "C:\Users\LabUser\Documents\bacula-windows-restore-test.txt" -Algorithm SHA256
```

Verified: Test-Path True; content “BACULA WINDOWS RESTORE TEST - 07 OCTOBER 2026”. Restored SHA256: 1B6AD84B07BB86A1F370DE413D7C5A9AA1DB7EFFBC3071230D0A0B5141715EC3. The two hashes captured after restoration match each other; this is not a comparison with a pre-deletion hash.

Run on BACULA-SERVER to check the restore job record:

```
sudo -u postgres psql -d bacula -c "SELECT jobid,name,jobstatus,jobfiles,jobbytes FROM job WHERE type='R' ORDER BY jobid DESC LIMIT 1;"
```

Observed JobId 56, RestoreFiles, T, 1 file, 372 bytes.

9. Nagios monitoring integration

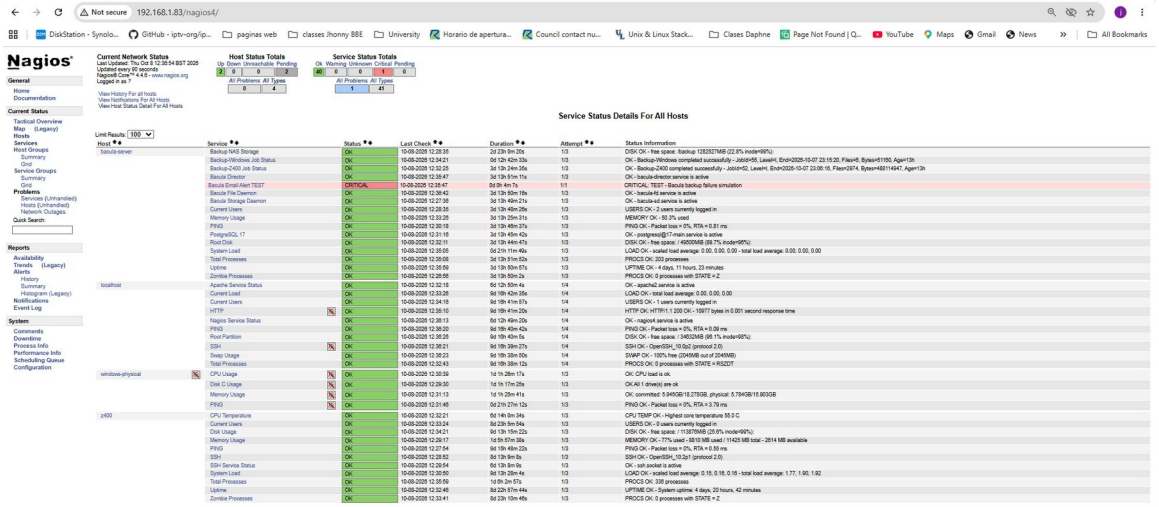


Figure 9. Nagios monitoring page during a deliberately simulated CRITICAL email-alert test; real Z400 and Windows backup checks remain OK.

On BACULA-SERVER, the existing /usr/lib/nagios/plugins/check_bacula_backup queries PostgreSQL as the nagios account, monitors Backup-Z400 and uses age thresholds of WARNING 30 hours / CRITICAL 48 hours. A copy was created for Windows and JOB changed to Backup-Windows, leaving the Z400 check intact.

Run on BACULA-SERVER:

```
cp -a /usr/lib/nagios/plugins/check_bacula_backup /usr/lib/nagios/plugins/check_bacula_windows_backup
sed -i 's/JOB="Backup-Z400"/JOB="Backup-Windows"/' /usr/lib/nagios/plugins/check_bacula_windows_backup
runuser -u nagios -- /usr/lib/nagios/plugins/check_bacula_windows_backup
```

Observed Windows plugin output: OK – Backup-Windows completed successfully – JobId 55, Level I, 6 files, 51,160 bytes. The script checks for a running job first, then inspects the latest completed

backup and reports non-T status as CRITICAL. Note: a job stuck indefinitely in Running would still report OK; production hardening should include maximum run duration.

NRPE command on BACULA-SERVER in /etc/nagios/nrpe.cfg:

```
command[check_bacula_windows_backup]=/usr/lib/nagios/plugins/check_bacula_windows_backup
```

On NAGIOS-SERVER (X.X.X.83), test remote execution:

```
/usr/lib/nagios/plugins/check_nrpe -H X.X.X.118 -c check_bacula_windows_backup
```

Observed: OK. Nagios service definition in /etc/nagios4/conf.d/bacula-server.cfg:

```
define service {
  use             generic-service
  host_name       bacula-server
  service_description Backup-Windows Job Status
  check_command     check_nrpe!check_bacula_windows_backup
}
```

Validation on NAGIOS-SERVER:

```
nagios4 -v /etc/nagios4/nagios.cfg
systemctl restart nagios4
```

The web UI displayed Backup-Z400 Job Status and Backup-Windows Job Status as OK. The wider monitoring environment included 40 operational services after removal of the temporary alert-test service.

10. Email alert validation

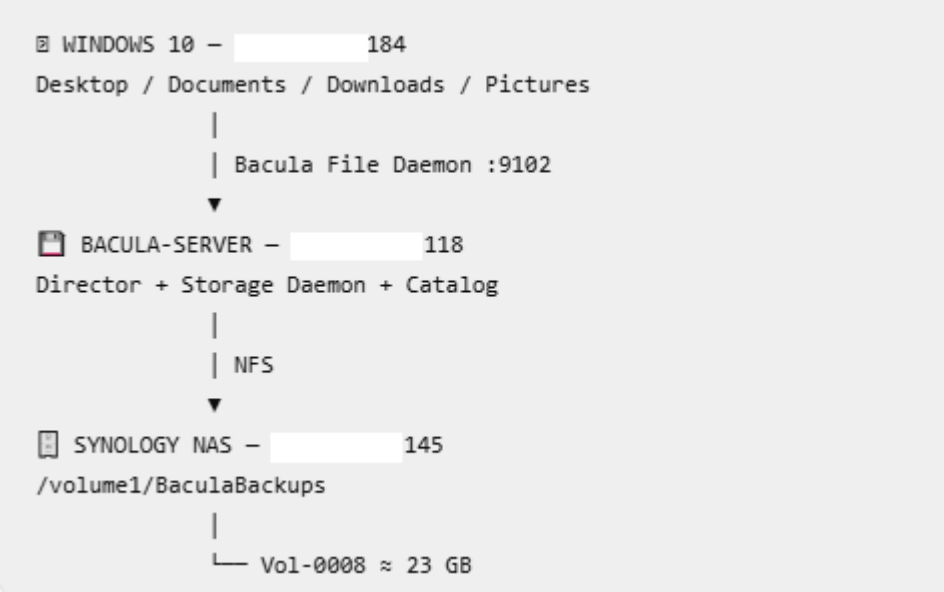


Figure 10. Windows–Bacula–Synology NAS data path, including Bacula FD port 9102 and NFS-backed storage.

Nagios generic-service inherited notifications_enabled=1, check_interval=10 minutes, retry_interval=2 minutes, max_check_attempts=3, notification_options=w,u,c,r,

notification_interval=60 minutes. The admins contact group includes nagiosadmin, routed to the configured Yahoo mailbox via Exim4. Exim4 was active.

A separate temporary Nagios service used check_dummy 2 to simulate CRITICAL, then check_dummy 0 to simulate OK. The CRITICAL email and subsequent recovery/OK email were both received. The temporary configuration /etc/nagios4/conf.d/bacula-alert-test.cfg was removed and disappeared from the web UI. This test did not interrupt real Bacula jobs.

11. Troubleshooting record

Symptom	Cause / resolution
Windows TCP 9102 unreachable	Windows Firewall blocked inbound FD port; add scoped rule allowing only X.X.X.118.
TLS illegal parameter / connection failure	Director name and shared FD password did not match Windows bacula-fd.conf; align settings.
bacula-dir: command not found	Use /usr/sbin/bacula-dir with explicit -c configuration path.
Restore path invalid	Override default Where with / (no prefix), restoring selected file to its original Windows location.
Nagios backup status absent	Add a distinct Windows NRPE command and Nagios service; validate configuration before restart.
Email alert validation	Use independent check_dummy service, test CRITICAL and OK, then remove temporary service.

12. Operational considerations and next improvements

Windows may enter sleep mode before the 23:05 scheduled job. Ensure wakefulness during the backup window or choose an appropriate schedule. Check NFS persistence and NAS free space; review Bacula retention, volume recycling, encrypted transport and off-site copy strategy. For stronger restore assurance, retain the original checksum before deletion and compare with the restored checksum. Consider monitoring excessive job run duration and adjusting backup freshness thresholds to reflect real schedules and maintenance windows.

13. Photographic and screenshot evidence

The screenshots included in this edition were supplied from the actual laboratory. They document the Windows Bacula installer, network checks, successful incremental jobs, restore workflow and Nagios alert simulation. Screenshots are historical evidence: the temporary CRITICAL test service shown in Figure 9 was subsequently removed after both CRITICAL and recovery emails were received. Other setup screenshots can be incorporated in a later edition if available.

14. Skills demonstrated

Linux and Windows administration; virtualisation; client/server backup architecture; PostgreSQL catalogue queries; NFS storage integration; TCP firewall troubleshooting; Bacula Full and

Incremental scheduling; file-level recovery; SHA256 integrity workflow; NRPE plugin integration; Nagios notifications; evidence-based validation and operational documentation.

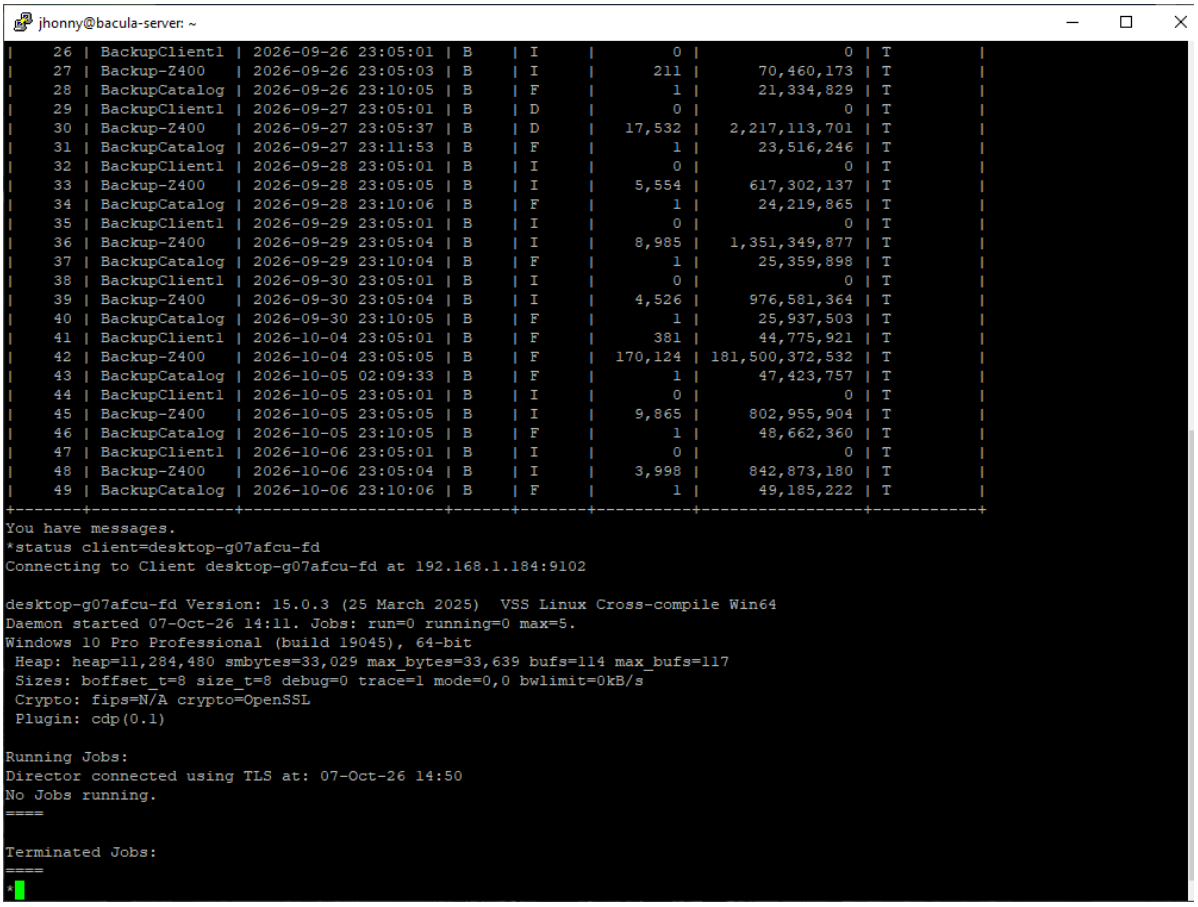
End of laboratory record. Prepared from verified lab outputs and implementation notes; sensitive passwords intentionally omitted.

15. Additional chronological evidence

The following additional screenshots supplement the installation, backup, recovery and Nagios alert procedures. Screenshots are evidence of the lab activities; where a screenshot is an intermediate state, it is not presented as proof of final success.

Figure 11. Windows File Daemon configuration and service diagnostics

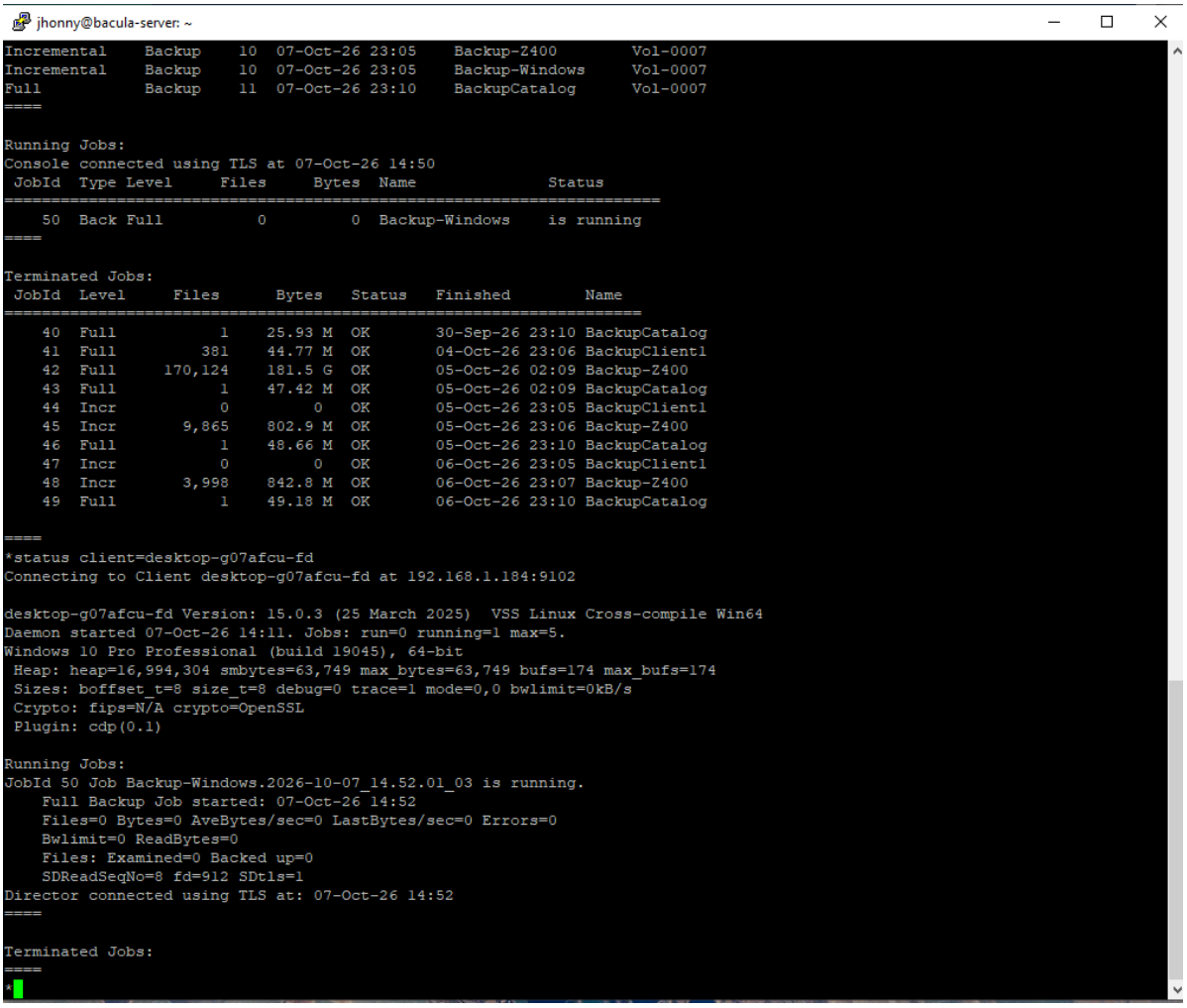
Windows 10 physical client — X.X.X.184. Inspect the service and client configuration; confirm the Director identity and TCP port 9102 before running backups.



Source: original laboratory screenshot supplied by the project owner.

Figure 12. Bacula job and storage inspection

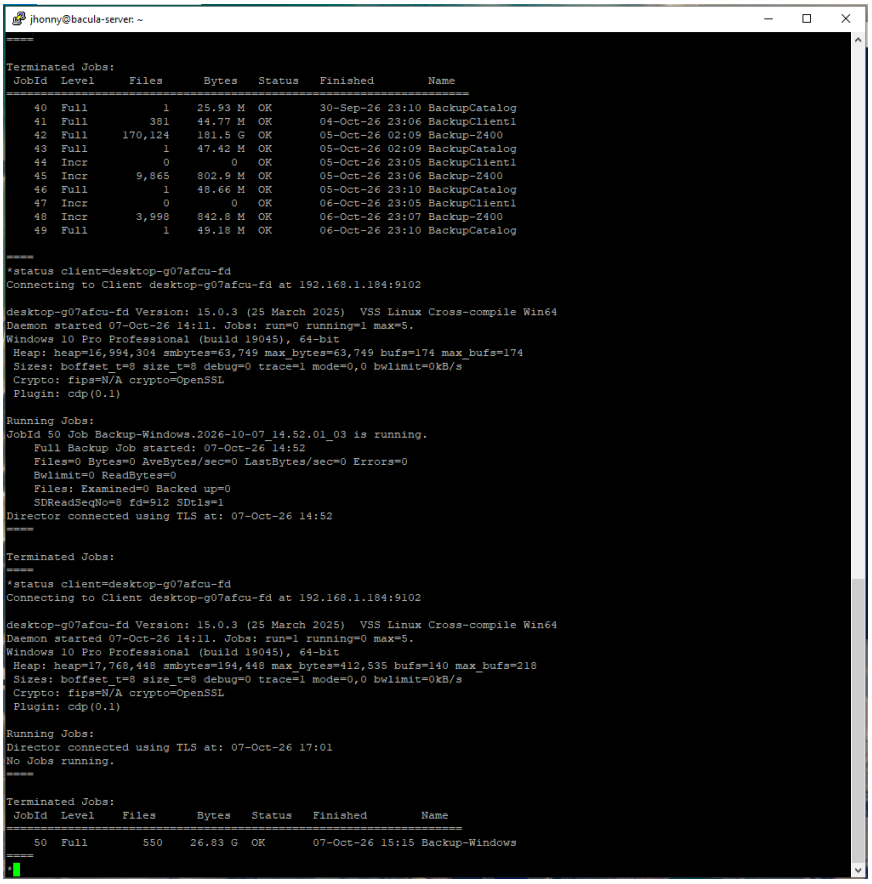
Bacula Server — X.X.X.118. Console output documenting configuration or job status during Windows integration.



Source: original laboratory screenshot supplied by the project owner.

Figure 13. Windows full backup completion evidence

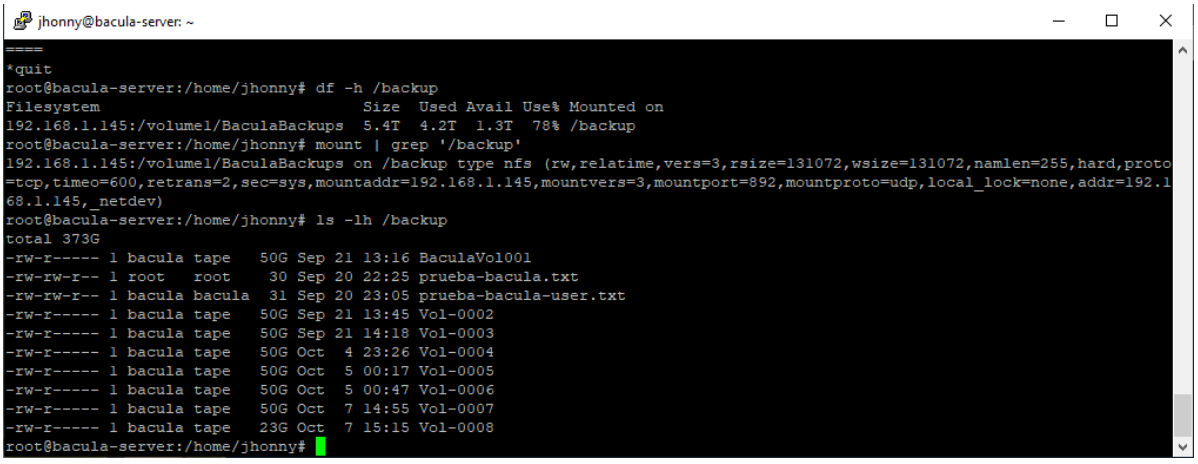
Bacula Server — X.X.X.118. Console output from the initial Windows backup phase; the validated full backup is JobId 50, 550 files, 26,833,571,540 bytes.



Source: original laboratory screenshot supplied by the project owner.

Figure 14. NAS volume and backup catalogue inspection

Bacula Server — X.X.X.118. Check the NFS-mounted backup directory and Bacula volumes; the observed Vol-0008 was approximately 23 GB after the Windows full backup.



Source: original laboratory screenshot supplied by the project owner.

Figure 15. Bacula recovery job selection

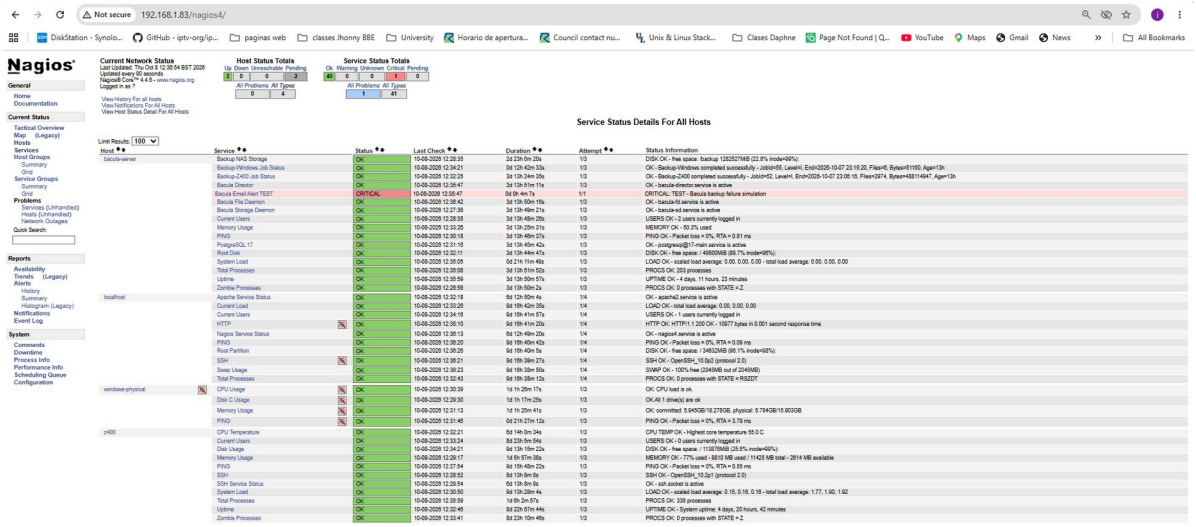
Bacula Server — X.X.X.118. bconsole restore workflow selecting Windows as the restore client and preparing a single-file recovery.

```
    13: Plugin Options
Select parameter to modify (1-13): 9
Please enter the full path prefix for restore (/ for none): /
Run Restore job
JobName:         RestoreFiles
Bootstrap:       /var/lib/bacula/bacula-server-dir.restore.1.bsr
Where:
Replace:         Always
FileSet:         Full Set
Backup Client:   bacula-server-fd
Restore Client:  desktop-g07afcu-fd
Storage:         File1
When:            2026-10-07 23:36:00
Catalog:         MyCatalog
Priority:        10
Plugin Options:  *None*
OK to run? (Yes/mod/no):
```

Source: original laboratory screenshot supplied by the project owner.

Figure 16. Nagios alert test: simulated CRITICAL

Nagios Server — X.X.X.83. The temporary Bacula Email Alert TEST appears in CRITICAL whilst the two production backup monitoring services remain OK.



Source: original laboratory screenshot supplied by the project owner.

Figure 17. Nagios email test configuration after recovery

Nagios Server — X.X.X.83. The previous CRITICAL check is commented out and the active check_dummy 0 command simulates recovery. The temporary service was subsequently removed.

```
jhonny@nagios-server: ~
root@nagios-server:/home/jhonny# cat /etc/nagios4/conf.d/bacula-alert-test.cfg
# Temporary Bacula email notification test

define command {
    command_name    check_bacula_alert_test
    #command_line    /usr/lib/nagios/plugins/check_dummy 2 "TEST - Bacula backup failure simulation"
    command_line    /usr/lib/nagios/plugins/check_dummy 0 "TEST - Bacula backup recovered successfully"


}

define service {
    use                     generic-service
    host_name               bacula-server
    service_description     Bacula Email Alert TEST
    check_command           check_bacula_alert_test
    max_check_attempts      1
    check_interval          1
    retry_interval          1
    notifications_enabled   1
    notification_options    c,r
    notification_interval   0
    notification_period     24x7
    contact_groups          admins
}
root@nagios-server:/home/jhonny#
```

Source: original laboratory screenshot supplied by the project owner.

16. Final verification and limitations

Windows RestoreFiles JobId 56 completed with status T, restoring one file (372 bytes). The recovered text file exists and contains: BACULA WINDOWS RESTORE TEST - 07 OCTOBER 2026.

Recovered-file SHA-256: 1B6AD84B07BB86A1F370DE413D7C5A9AA1DB7EFFBC3071230D0A0B5141715EC3. A matching pre-deletion checksum was not available in the supplied evidence, so byte-for-byte cryptographic equivalence with the original has not been independently established.

Nagios notification validation: one CRITICAL test email and a second recovery/OK email were received. The temporary Bacula Email Alert TEST service was removed afterwards. The production Backup-Z400 Job Status and Backup-Windows Job Status services remained operational.